

## Research Article · Computation &amp; Mathematics

# Unpredictable by Design: Entropy-Regularised Allocation for Multi-Threat Wildlife Protection

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**Background.** Etosha National Park in Namibia covers 22,935 km<sup>2</sup> and faces three threats at once. It lost 46 black rhinos to poaching in 2022 [1]; 34% of its area burned in a single week in September 2025; and elephants breach its 850 km perimeter fence continually. About 100 rangers are on duty per shift. Every ranger sent to one threat is unavailable for the other two, which makes protection a three-way allocation problem rather than three separate problems.

**Methods.** We represent the park two ways at once — a weighted road network joining 86 waterholes, and a 2.5 km terrain grid — because poaching is a discrete site problem and fire is a continuous spread problem. A Conservation Priority Index and a Habitat Integrity Index score what is worth protecting. Three threat models run concurrently: poaching from a log-linear rate model, fire from a time-squared damage index, and fence breaches from animal pressure and fence condition. Protection is treated as a probability per site rather than a binary label, and aggregated into a Park Protection Score (PPS) on [0, 1].

**Findings.** A deterministic schedule that maximises expected protection leaves 63 of 86 waterholes at zero coverage every shift — a fixed pattern any adversary with an informant can learn. We replace it with Entropy-Regularised Patrol Optimisation (ERPO), a soft-max allocation that trades expected damage prevented against schedule unpredictability. Marginal Protection Value coordinate descent then splits rangers across the three threats, converging at 69/8/23. With 30 cameras and 5 drones the framework reaches PPS = 0.545, 54% above baseline. ERPO costs 26.6% against a perfectly uninformed adversary but overtakes deterministic patrol once informant accuracy exceeds  $\epsilon \approx 0.19$ . Monte Carlo simulation over 200 replications confirms the predictions within  $\pm 5\%$ , and single-parameter sensitivity moves PPS by less than 2%.

**Limitations.** The model has never been run against a real patrol schedule. Species densities come from regional proxies rather than direct survey, the poaching rate model is built from ecological reasoning rather than fitted to incident data (which is classified), and PPS is a constructed index whose absolute value has no external referent.

## 1. The problem

Conservation planning usually treats threats one at a time. A park has an anti-poaching plan, a fire plan, and a human-wildlife conflict plan, each written as though the others did not compete for the same people. That works while staffing is comfortable. It stops working at Etosha [2], where roughly 100 rangers are on duty at any moment across an area the size of Wales, and where all three threats are active simultaneously (Figure 1).

The three are also geometrically different, which is why a single model rarely fits all of them. Poaching happens at *places*: animals concentrate at waterholes, and so do poachers, so the problem is discrete and lives on a network. Fire is a *process*: it spreads across terrain regardless of where

anything is placed, so the problem is continuous and lives on a grid. Fence breaches happen along a *line*: 850 km of perimeter with varying condition and varying animal pressure behind it.

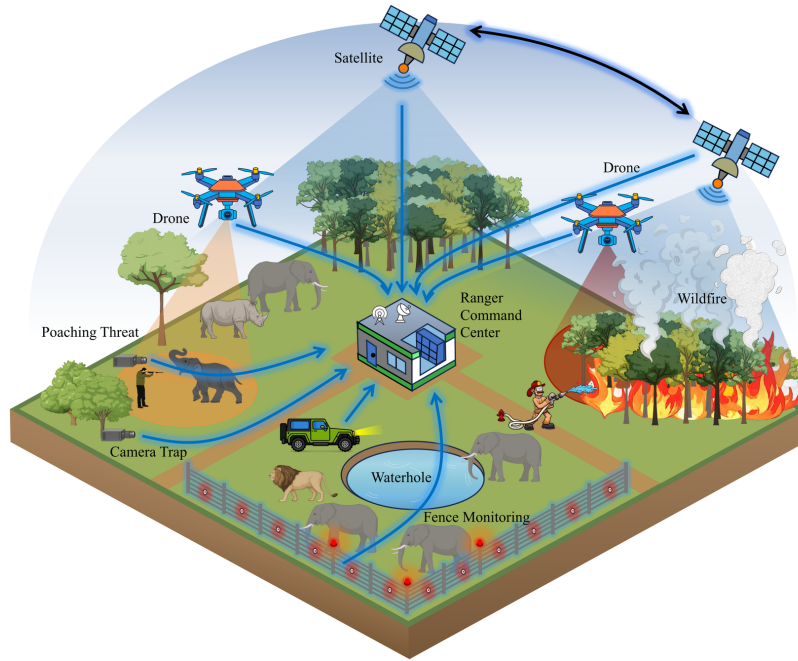
Forcing these into one abstraction loses information. Our first design decision was therefore to keep two spatial representations and match each threat to the geometry it actually has (Figure 2).

## 2. What is worth protecting

Allocation requires a value to allocate against. We construct two indices.

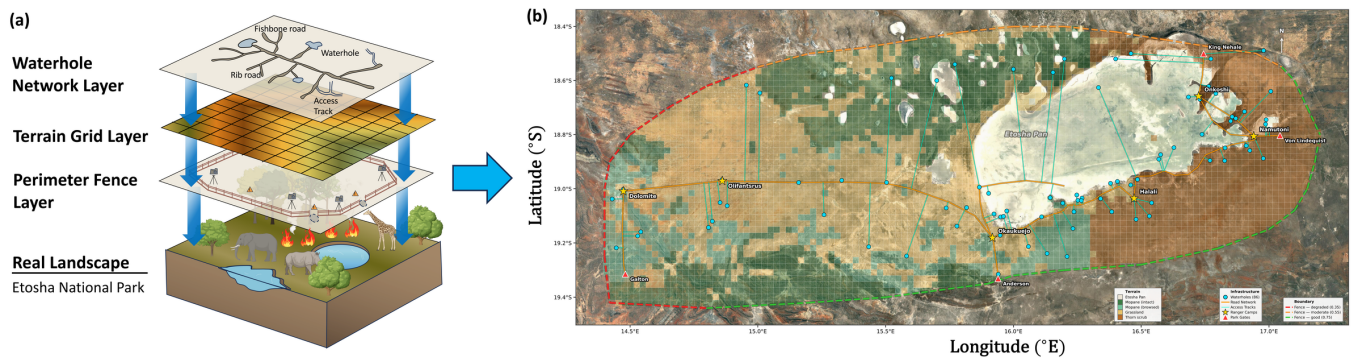
The **Conservation Priority Index** (CPI) scores each of the 86 waterholes by the species present, their conservation status, and their density in the surrounding area [3, 4]. A

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**Figure 1. What the allocation problem looks like on the ground.** Detection comes from camera traps, drone sorties and satellite passes; response comes from rangers who can only be in one place at a time.

All three threats draw on the same command centre and the same people. That shared pool is what makes this one allocation problem rather than three plans.



**Figure 2. Two representations of one park.** (a) The park is modelled as stacked layers: a waterhole network for poaching, a 2.5 km terrain grid for fire, and the perimeter fence as a line.

(b) The same layers over Etosha: 86 waterholes on the road network, the terrain grid coloured by vegetation class, and fence segments coloured by condition from degraded (0.35) to good (0.75). Poaching lives on the points, fire on the cells, breaches on the line.

waterhole used by black rhino scores higher than one used only by common ungulates, because the loss is less recoverable — an irreversibility premium rather than a simple headcount.

The **Habitat Integrity Index** (HII) scores each 2.5 km terrain cell by vegetation type (from the ESA WorldCover layer [5]), fuel load, and its contribution to habitat continuity. This is what fire destroys, and it is not the same quantity as the animals at a waterhole.

### 3. Three threats, three models

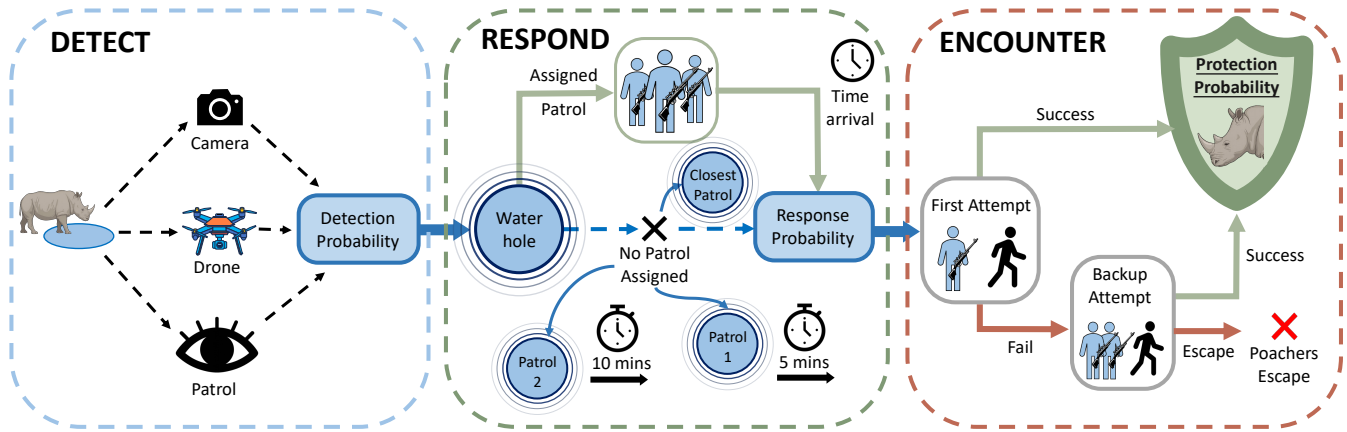
#### 3.1 Poaching

Poaching pressure at a waterhole is modelled as a log-linear function of the value present, its accessibility from outside the park, and its distance from ranger infrastructure. The form follows the incentive models of Milner-Gulland and Leader-Williams [6, 7], and the clustering of repeat incidents echoes the self-exciting point processes used for

crime [8]. Rhino numbers and translocation history for Etosha come from [9]. High-value sites that are easy to reach and far from a camp carry the highest rate.

#### 3.2 Fire

Fire damage is not linear in response time. A fire spreads roughly as the square of the delay before it is reached, so the Fire Damage Index is taken as  $FDI \propto \tau^2$ ; the quadratic follows from grassland spread experiments [10, 11], and September 2025 provided the case that motivated it [12–14]. This single choice drives the technology case: at the modelled spread rate, a drone reaching a fire in 1 hour and a satellite detection arriving at 9 hours differ by a factor of about 81 in area burned; the nine-hour figure is the MODIS revisit interval [15], and thermal-sensor drones are already used for exactly this [16, 17]. Detection latency, not suppression capacity, is the binding constraint.



**Figure 3. How a coverage probability is built.** Protection at a waterhole is not a single number but the product of three stages. A threat must be **detected** — by camera, drone or patrol; a patrol must **respond** — either one already assigned,

or the nearest free one, arriving late; and the **encounter** must succeed, with one backup attempt. Each stage can fail independently, which is why coverage is continuous rather than binary.

### 3.3 Human–wildlife conflict

Breach rate along each fence segment is modelled from the animal pressure behind it — largely elephant density, which varies seasonally with water availability — and the physical condition of that segment. Fire and herbivory interact in savanna in ways that make the two threats hard to separate [18]. Travel times on the road network use posted park speed limits [19], and camera detection probabilities follow [20].

### 3.4 Protection as a probability

With 86 waterholes and far fewer patrol teams, no schedule can call a site “protected” or “unprotected” honestly. Each site instead carries a continuous coverage probability — itself the product of detection, response and encounter, as Figure 3 sets out — and these aggregate into a composite **Park Protection Score**:

$$\text{PPS} = \omega \Pi_S + (1 - \omega - \gamma) \Pi_H + \gamma p_{\text{prev}}$$

where  $\Pi_S$  is the anti-poaching score,  $\Pi_H$  the habitat score, and  $p_{\text{prev}}$  the breach-prevention score. PPS lies in  $[0, 1]$  and is the single quantity the allocation maximises.

### 4. Why a good schedule is a bad schedule

The obvious allocation is greedy: rank waterholes by expected damage prevented and patrol the top ones. It maximises expected protection, and on our model it lifts  $\Pi_S$  from 0.129 to 0.271 — the single largest gain of any step.

It also leaves **63 of 86 waterholes at zero coverage in every shift**, and the same 63 every time. That is not a rounding artefact; it is what optimality looks like when resources are scarce. And it is a schedule an adversary can learn. Rangers are local, and informants are not hypothetical: Moreto’s fieldwork in Uganda documents ranger wrongdoing including the sale of patrol information [21]. A poacher who knows which sites are never visited faces no detection risk at all.

The opposite extreme — patrol uniformly at random — is unexploitable but ignores that waterholes differ enormously in value. **Entropy-Regularised Patrol Optimisation** interpolates between them. The idea is not new — Xu

et al. add an entropy term to a security game for the same reason [22], and PAWS deploys Stackelberg patrol planning in real parks [23]. What we add is the coupling to two other threats competing for the same rangers, which neither addresses. Patrol probability at site  $i$  is a softmax over damage weight  $m_i$  with temperature  $\mu$ :

$$c_i^* \propto \exp(m_i/\mu)$$

Small  $\mu$  approaches greedy; large  $\mu$  approaches uniform. The entropy term guarantees every site retains non-zero probability, so no site is permanently safe to raid. We take  $\mu^* = 0.93$ .

### 5. Splitting rangers across threats

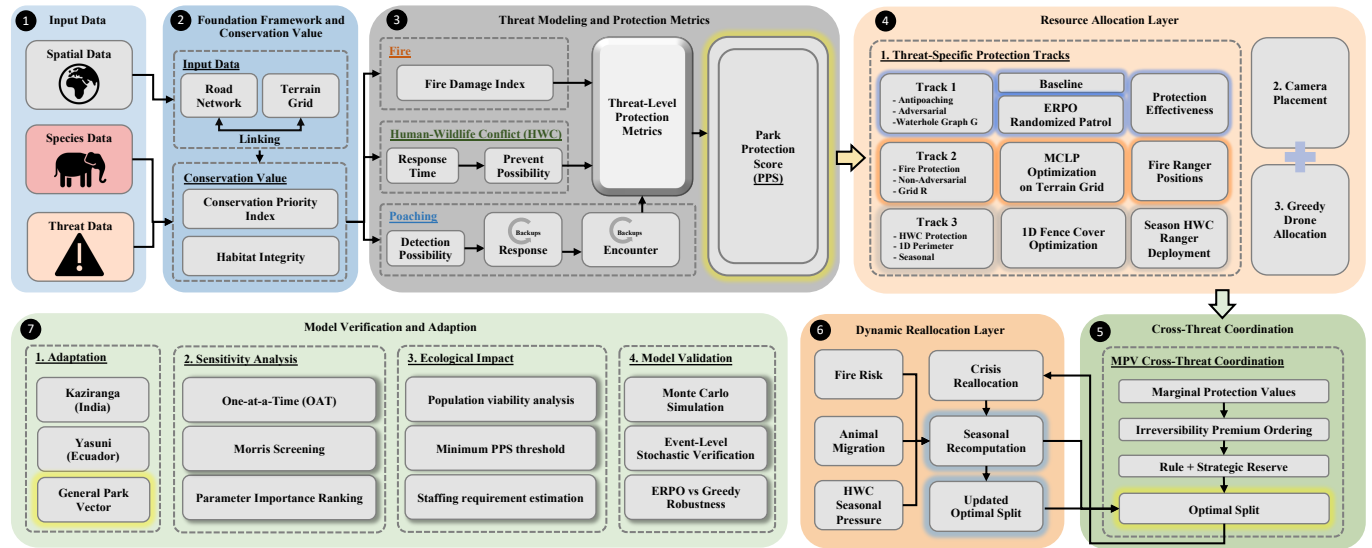
Figure 4 shows how the pieces fit together end to end. Each threat pool is optimised internally, but the pools must also be sized against each other. We use **Marginal Protection Value** coordinate descent: move one ranger from the pool with the lowest marginal PPS gain to the pool with the highest, and repeat until they equalise. Fire ranger placement is a maximal covering location problem [24], and its greedy solution inherits the  $1 - 1/e$  guarantee for monotone submodular objectives [25]. The allocation converges at **69 anti-poaching / 8 fire / 23 HWC**.

The small fire allocation is not a judgement that fire matters less. It follows from Figure 5(c): because damage grows as  $\tau^2$ , the return on fire response comes almost entirely from detection speed, which drones and cameras supply more cheaply than rangers do. Staffing fire heavily would buy little.

A **crisis reallocation** rule overrides the converged split temporarily when an acute event makes one threat’s marginal value spike — an active fire, or a confirmed incursion. The split is a steady state, not a rule to follow through an emergency.

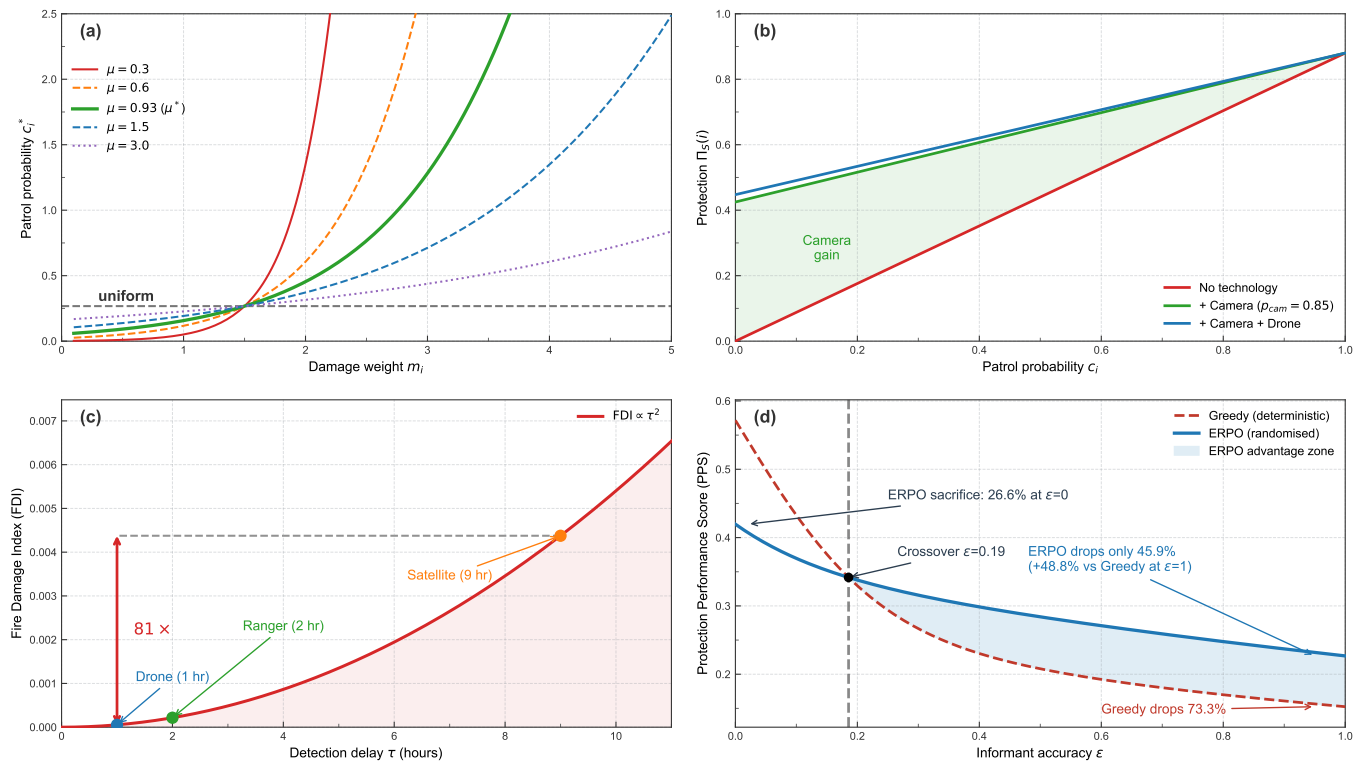
### 6. Results

Table 1 sets out what each step of the pipeline buys, and Figure 6 maps the deployment those numbers describe. Two rows deserve comment. ERPO *lowers*  $\Pi_S$  from 0.271 to



**Figure 4. The whole framework, end to end.** 1–2 Spatial, species and threat data feed the two representations and the two value indices. 3 Each threat is modelled on its own domain and reduced to a protection metric, which aggregate into the Park Protection Score. 4 Each threat gets its own allocation track — ERPO for poaching, maximal-covering optimisation on the terrain grid for fire, one-dimensional

cover optimisation along the fence for HWC — with camera and drone placement alongside. 5 Marginal Protection Values coordinate the three pools into one split. 6 Seasonal and crisis triggers recompute that split when conditions move. 7 Sensitivity, ecological-impact and Monte Carlo checks, and transfer to other parks through a single characterisation vector.



**Figure 5. The four relationships the model turns on.** (a) Temperature  $\mu$  interpolates between greedy and uniform patrol. (b) Cameras raise protection most where patrol probability is lowest — they substitute for presence, not for effort. (c) Fire damage grows as the square of detection delay; the drone-to-satellite gap is a factor of 81.

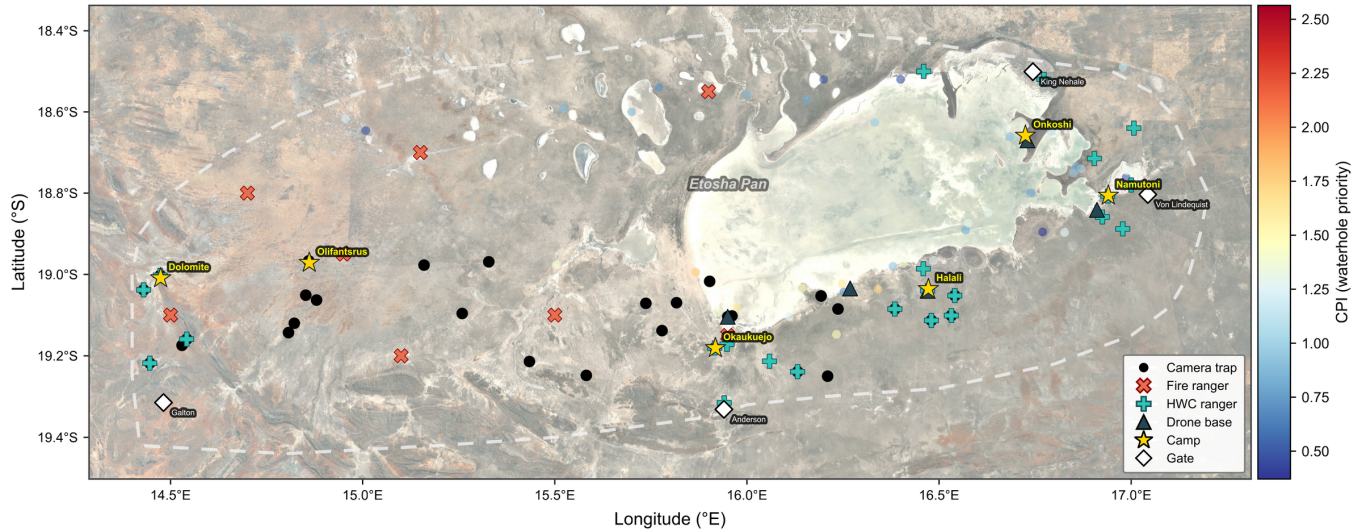
(d) The central result. Against a perfectly uninformed adversary ( $\epsilon = 0$ ) greedy patrol wins and ERPO gives up 26.6%. The lines cross at  $\epsilon \approx 0.19$ . Beyond it greedy collapses by 73.3% while ERPO loses only 45.9%, because a randomised schedule cannot be sold.

0.126 — the cost of unpredictability, paid in expected protection. That trade is only worth making if the adversary is informed, which is the question Figure 5(d) answers. And the HWC step is the largest single jump in PPS, because the baseline leaves fence prevention at zero: the cheapest gains

were never in the threat that gets the most attention.

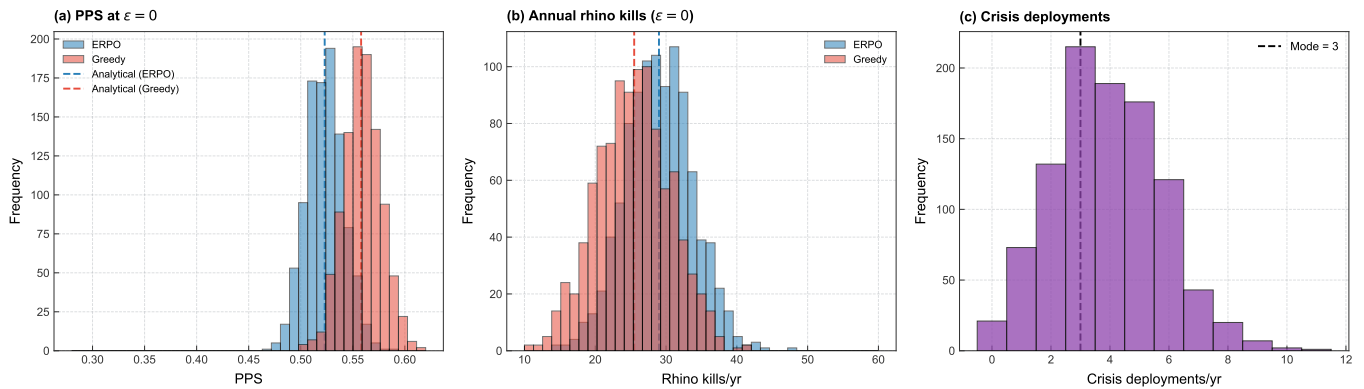
## 7. Robustness

Single-parameter sensitivity moves PPS by less than 2% across the tested range for every threat-specific parameter, so the deployment is not balanced on a knife edge of as-



**Figure 6. The resulting deployment.** Camera traps cluster on the high-CPI waterholes south and west of the pan; HWC rangers concentrate along the eastern and north-eastern fence where elephant pressure is highest;

fire rangers spread across the western fuel-load gradient. Three drone bases cover the park from Namutoni, Halali and the central pan.



**Figure 7. Stochastic verification.** 200 replications against the analytic model.

Agreement within  $\pm 5\%$  across configurations, with ERPO showing a narrower outcome distribution than greedy patrol.

| Configuration                    | $\bar{\Pi}_S$ | $\bar{\Pi}_H$ | $\bar{p}_{prev}$ | PPS          |
|----------------------------------|---------------|---------------|------------------|--------------|
| Baseline (proportional, no tech) | 0.056         | 0.933         | 0.000            | 0.355        |
| + 30 cameras                     | 0.129         | —             | 0.000            | 0.391        |
| + Greedy deterministic patrol    | 0.271         | —             | —                | 0.462        |
| + ERPO (randomised, $\mu^*$ )    | 0.126         | —             | —                | 0.390        |
| + Fire ranger placement          | —             | 0.941         | —                | 0.392        |
| + HWC rangers on fence           | —             | —             | 0.856            | 0.521        |
| + 5 drones                       | 0.126         | 0.941         | —                | 0.522        |
| <b>Full model, dry (69/8/23)</b> | <b>0.164</b>  | <b>0.939</b>  | <b>0.898</b>     | <b>0.545</b> |
| <b>Full model, wet (69/8/23)</b> | <b>0.164</b>  | <b>0.939</b>  | <b>0.898</b>     | <b>0.560</b> |

**Table 1.** Progressive results. The gap between rows is the marginal contribution of each step.

summed constants. The result is sensitive to the weights  $\omega$  and  $\gamma$  in the PPS definition, which is expected: those encode a value judgement about how to trade rhinos against habitat, and no amount of modelling can settle that.

Monte Carlo simulation over 200 replications reproduces the analytic predictions within  $\pm 5\%$  (Figure 7). It also shows something the analytic model cannot: ERPO reduces the *variance* of outcomes, not only their exploitability. A deterministic schedule is either right or wrong about where the adversary goes; a randomised one is partly right

more often.

## 8. What sustainability would actually require

PPS = 0.545 is a 54% improvement, and it is not enough. Fitting a logistic harvest model to the protected population, with the minimum viable population literature setting the floor we test against [26], gives a minimum viable protection threshold of  $\text{PPS} \geq 0.94$  — below which the modelled population declines rather than recovers.

Reaching it requires approximately **1,050 staff**, about  $3.6\times$  current levels. We report this because the honest output of an allocation model is not only the best use of what exists but the distance between that and what the problem needs. Optimising 100 rangers well does not make 100 rangers sufficient.

## 9. Transfer to other parks

The architecture takes a single park characterisation vector — area, perimeter, waterhole count, staffing, threat weights — rather than being fitted to Etosha. Applying it unchanged gives PPS = 0.67 for Kaziranga (India) and PPS = 0.54 for Yasuní (Ecuador). We present these as a demon-

stration that the pipeline runs on other inputs, not as validated results for those parks.

## 10. Limitations

- **No ground truth.** The model has never been compared against a real patrol schedule and real outcomes. Every number here is internal to the model.
- **Species densities are proxies.** Most come from regional estimates rather than direct survey at each waterhole, and that uncertainty propagates into every downstream allocation. Entropy regularisation and sensitivity testing are partial insurance; systematic mischaracterisation of individual waterholes remains possible.
- **The poaching rate model is reasoned, not fitted.** Historical incident data for Etosha is classified, so the log-linear form rests on ecological argument rather than on observation.
- **PPS has no external referent.** It is a constructed index. “0.545” is meaningful only relative to other configurations of the same model, and the 0.94 threshold inherits the assumptions of the harvest model it came from.
- **The adversary is a parameter.** Informant accuracy  $\varepsilon$  is the quantity the central result turns on, and we do not measure it — we show where the crossover lies and leave the reader to judge which side of  $\varepsilon \approx 0.19$  a real park sits on.

## 11. Conclusion

Treating poaching, fire and fence breaches as one allocation problem rather than three plans changes what the optimum looks like. Two spatial representations preserve each threat’s geometry; probabilistic coverage replaces a binary that scarce resources cannot support; and entropy regularisation buys robustness against an informed adversary at a measurable price in expected protection. Whether that price is worth paying is not a modelling question but an empirical one, and it reduces to a single number: how much the other side already knows.

## Data and code availability

Deposited together at <https://doi.org/10.5281/zenodo.21847180> (CC BY 4.0): park geometry, the 86 waterholes with coordinates, road network, fence segments, species densities with sources cited per value, the 2.5 km terrain grid, and the full analysis pipeline — indices, the three threat models, ERPO, ranger placement, MPV coordinate descent and the Monte Carlo simulation — with one script per figure. Every figure regenerates from the pipeline; panel (d) of Figure 5, on which the central claim rests, comes from `figures/figure_analytical_quartet.py`. Etosha’s historical poaching incident records are classified and are not part of the deposit.

## Competing interests

Three of the four authors hold editorial roles at PRISM: Henry Wang is Editor-in-Chief and Aiden Song and Alvin Hsiang are Deputy Editors. All three recused themselves entirely from the handling of this manuscript, which was managed by a deputy editor with no involvement in the work and reviewed by referees who are not authors. This declaration is published with the article.

## Mentor contribution

Cloud Wu, our coach, gave us a deep analysis of the real conservation problem this work addresses, and kept the modelling from drifting away from the question it set out to answer. That framing shaped what we chose to measure. They did not derive the models, write the code, produce the figures, or write any part of this manuscript.

## Use of AI tools

**During the modelling work (KIMI K2.5, March 2026).** Draft paragraphs in the introduction, threat modelling and resource allocation sections were polished for clarity and academic tone; every output was edited for technical accuracy and checked against our cited sources before use. Starter code was generated for three components — the Monte Carlo replication loop, the ERPO softmax with log-sum-exp stabilisation, and the MPV coordinate descent — and each was then substantially rewritten by us: the 86-waterhole network topology, the three-stage detection pipeline, seasonal parameter updates, crisis reallocation, the unmet-need weighting, integer constraints, the irreversibility scaling and multi-start initialisation are our own. Matplotlib scaffolding for several figures was generated and then rebuilt with the real data and projections.

**Running and debugging the analysis (Claude Code, Anthropic, March 2026).** One team member used Claude Code on their own machine to execute the figure scripts, install the geospatial dependencies, and work through plotting errors during the competition period. It was used to run and debug code, not to design the models or generate results.

**In preparing this manuscript (Claude, Anthropic, 2026).** The competition report was rewritten into this article: the prose was drafted from our  $\text{\LaTeX}$  source, figures were selected, and the typeset PDF produced. All models, data, analysis, results and figures are our own work from the original submission, and every numerical claim was checked against the source and against the deposited code.

No result in this paper was produced by a generative tool. Disclosed use is never held against a manuscript; undisclosed use is a research-integrity matter, so we have listed everything.

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